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Bihar Engineering University, Patna

B.Tech 7th Semester Exam., 2021

Code: 100708 ***Time: 03 Hours***

Semester: VII ***Subject: Biology for Engineers (New Course)*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven): **[2 x 7 = 14]**

(a) The digestive system of the cell is known as
(i) lysosome (iii) ribosome
(ii) chloroplast (iv) Golgi body

(b) An organism carrying two identical alleles is
(i) recessive (iii) homozygous
(ii) heterozygous (iv) dominant

(c) Which allows the body to produce vitamin D?
(i) Calcium (iii) Steroid
(ii) Cholesterol (iv) Acetone

(d) The longest part of the cell cycle is
(i) prophase (iii) interphase
(ii) metaphase (iv) telophase

(e) 'ABO' system of blood measures the
(i) protein (iii) antibody
(ii) sugar (iv) antigen

(f) Study of birds is called
(i) haematology (iii) saurology
(ii) ornithology (iv) rhinology

(g) The term 'biology' was introduced by
(i) Lamarck and Traviranus (iii) John Ray
(ii) Darwin (iv) Aristotle

(h) Gases have
(i) low density (iii) brutality
(ii) ductility (iv) high density

(i) Movement of particles in liquids and gases is observed as
(i) Bruneian motion (iii) Blackian motion
(ii) Brownian motion (iv) Randomium motion

(j) Thermodynamics does not consider
(i) initial and final state (iii) bulk of matter
(ii) time factor (iv) energy change

Q.2 Write the answers in very brief : **[3½ x 4 = 14]**

(a) Write the basic working principle of data storage in human brain.

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(b) Define cell division.

(c) Write the concepts of recessive and dominance of gene.

(d) What are the basic properties of a model organism?

Q.3 Write the answers in very brief : **[3½ x 4 = 14]**

(a) Write the difference between unicellular and multicellular organisms.

(b) Differentiate between plant cell and animal cell.

(c) List the laws of inheritance.

(d) Define molecular taxonomy of living organism.

Q.4 Write the answers in short : **[7 x 2 = 14]**

(a) Write in brief properties and examples of all three major kingdoms of life.

(b) Explain the mechanism of ammonia excretion from human body.

Q.5 Write the answers in short : **[7 x 2 = 14]**

(a) Differentiate between autotroph and heterotroph organisms.

(b) Write the comment on model organisms C. elegans and A. thaliana.

Q.6 Write the answers in short : **[7 x 2 = 14]**

(a) Discuss about discovery of 20th Century that leads to major investigation in animal science.

(b) Describe the mechanism of mitosis cell division.

Q.7 Write the answers in short : **[7 x 2 = 14]**

(a) Discuss about law of dominance, law of segregation and law of independent assortment of Mendel.

(b) Differentiate between gene interaction and epistasis.

Q.8 Write the answer in very long : **[14]**

Draw a well-labelled diagram of animal and plant cell and list specific functions of each cell organ.

Q.9 Write the answer in very long : **[14]**

What is modern biology? Distinguish between traditional and modern biology.

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